

Proof. Let $a \in A$. Choose some simple number $n > 1$ with $a^n = a$. Let c be the characteristic of A , which satisfies $c > 0$ (since $2^m = 2$ for some $m > 0$). The ring $\mathbb{Z}[a] = (\mathbb{Z}/c\mathbb{Z})[a]$ is finite, because it is generated by $1, \dots, a^{n-1}$ as a $\mathbb{Z}/c\mathbb{Z}$ -module. By applying the Chinese Remainder Theorem to the finite reduced commutative ring $\mathbb{Z}[a]$, we can write $\mathbb{Z}[a]$ as a finite direct product of finite fields. If $\mathbb{F}_q$ is one of these fields, choose some generator ζ of its multiplicative group and some simple number $m > 1$ such that $\zeta^m = \zeta$. Then $\zeta^{m-1} = 1$, so that $q-1 \mid m-1$. Since m is simple, q is a prime number. Therefore, $\mathbb{Z}[a]$ is actually a finite direct product of prime fields. If $e_1, \dots, e_s \in \mathbb{Z}[a]$ are the corresponding idempotent elements, it follows that a is a $\mathbb{Z}$ -linear combination of $e_1, \dots, e_s$. Hence, a is central by Lemma 2.8. $\square$

In particular, we have a short proof that for simple n every n -ring is commutative. In fact, n is simple if and only if every n -ring is additively generated by idempotents. The goal of this section is to make this proof even more constructive. Because of Theorem 3.14, it suffices to look at n -rings with $p = 0$. We look at an example first.

Example 5.3. The free 5-ring with $5 = 0$ on one generator is $\mathbb{F}_5[T]/(T^5 - T)$. First we will write $[T]$ as a linear combination of idempotent (and hence central) elements. The Chinese Remainder Theorem implies that

$$\mathbb{F}_5[T]/(T^5 - T) \rightarrow (\mathbb{F}_5)^5, \quad [f] \mapsto (f(0), f(1), f(2), f(3), f(4))$$

is an isomorphism. The image of $[T]$ is $(0, 1, 2, 3, 4)$, which is the integer linear combination $1 \cdot (0, 1, 0, 0, 0) + 2 \cdot (0, 0, 1, 0, 0) + 3 \cdot (0, 0, 0, 1, 0) + 4 \cdot (0, 0, 0, 0, 1)$ of the canonical idempotents. Their preimages are the classes of

$$e_i(T) := 1 - (T - i)^4 \in \mathbb{F}_5[T]$$

for $0 \leq i \leq 4$. This is because for $u \in \mathbb{F}_5$ we have $e_i(u) = 1$ for $u = i$ and $e_i(u) = 0$ for $u \neq i$. Hence,

$$[T] = 1 \cdot [e_1] + 2 \cdot [e_2] + 3 \cdot [e_3] + 4 \cdot [e_4].$$

This leads to the following equational proof that 5-rings with $5 = 0$ are commutative: Let x be any element. A calculation (using only $5 = 0$) shows that

$$1 \cdot (1 - (x - 1)^4) + 2 \cdot (1 - (x - 2)^4) + 3 \cdot (1 - (x - 3)^4) + 4 \cdot (1 - (x - 4)^4) = x,$$

or equivalently

$$4 \cdot (x - 1)^4 + 3 \cdot (x - 2)^4 + 2 \cdot (x - 3)^4 + 1 \cdot (x - 4)^4 = x.$$

Since each y^4 is central by Lemmas 2.8 and 2.9, it follows that x is central.

We will now generalize this example.

Lemma 5.4. *Let q be a prime power. Let A be an $\mathbb{F}_q$ -algebra and $x \in A$. Then there is an equational proof of*

$$x = \sum_{u \in \mathbb{F}_q^\times} u \cdot (1 - (x - u)^{q-1}). \quad (4)$$

The equational proof takes place in the theory of $\mathbb{F}_q$ -algebras. This means that the arithmetic of $\mathbb{F}_q$ is taken for granted.

Proof. The abstract proof of eq. (4) works as follows: It is enough to verify the equation

$$X = \sum_{u \in \mathbb{F}_q^\times} u \cdot (1 - (X - u)^{q-1})$$

in the universal example $\mathbb{F}_q[X]$. Both sides are polynomials of degree $< q$ in X , so that it suffices to prove that the corresponding polynomial functions $\mathbb{F}_q \rightarrow \mathbb{F}_q$ agree. This follows from the observation that for $\alpha \in \mathbb{F}_q$ we have $(\alpha - u)^{q-1} = \delta_{u,\alpha}$.

The equational proof looks as follows: The case $q = 2$ is easy to check. Now let $q > 2$. We first prove that for all $j \in \mathbb{Z}$ we have

$$\sum_{u \in \mathbb{F}_q^\times} u^j = \begin{cases} -1, & q-1 \mid j, \\ 0, & \text{else} \end{cases} \quad (5)$$

in $\mathbb{F}_q$. The proof is well-known. Choose a generator ζ of $\mathbb{F}_q^\times$, and let s be the sum in eq. (5). Then $\zeta^j s = s$. If $\zeta^j \neq 1$, we get $s = 0$. Otherwise, we have $q-1 \mid j$ and then each summand of s is 1, so that $s = q-1 = -1$. This proves eq. (5). For $j = 1$ we get in particular $\sum_{u \in \mathbb{F}_q^\times} u = 0$ (here, we use $q > 2$).

Now, the binomial theorem and eq. (5) give us

$$\begin{aligned} \sum_{u \in \mathbb{F}_q^\times} u \cdot (1 - (x - u)^{q-1}) &= \sum_{u \in \mathbb{F}_q^\times} u - \sum_{u \in \mathbb{F}_q^\times} \sum_{i=0}^{q-1} \binom{q-1}{i} \cdot u \cdot (-u)^{q-1-i} \cdot x^i \\ &= - \sum_{i=0}^{q-1} \binom{q-1}{i} (-1)^{q-1-i} \cdot \left(\sum_{u \in \mathbb{F}_q^\times} u^{q-i} \right) \cdot x^i \\ &= - \binom{q-1}{1} (-1)^{q-1-1} (-1) \cdot x = x. \end{aligned} \quad \square$$

Theorem 5.5. *Let q be a prime power. Let A be an $\mathbb{F}_q$ -algebra and $x \in A$. If $x^q = x$, then x can be explicitly written as an $\mathbb{F}_q$ -linear combination of idempotent elements as in eq. (4). In particular, if A is a q -ring, A is commutative by an equational proof.*

Proof. We apply Lemma 5.4. If $u \in \mathbb{F}_q$, then $(x - u)^q = x^q - u^q = x - u$ since x, u commute, and then $(x - u)^{q-1}$ is idempotent by Lemma 2.9. Then x is central by Lemma 2.8. $\square$

Remark 5.6. A different equational proof of the commutativity of p -rings with $p = 0$ has been found by Forsythe and McCoy [FM46, Section 3]. It uses the Vandermonde determinant and is similar to our proof for p^2 -rings below (Theorem 6.4).

Corollary 5.7. *For every simple number $n > 1$ there is an equational proof that every n -ring is commutative.*

Proof. By Theorem 3.14 it is enough to find an equational proof that n -rings with $p = 0$ are commutative, where $p \in \mathbb{P}$ satisfies $p-1 \mid n-1$. By Theorem 4.1 and the definition of a simple number every such ring is actually a p -ring by an equational proof. Now Theorem 5.5 finishes the proof. $\square$

Let us check how the equational proof looks like in some examples.

Example 5.8. Consider $n = 3$. First we will use some non-equational concepts, but we will get rid of them afterwards. Let A be a 3-ring. Then $0 = 2^3 - 2 = 6 = 2 \cdot 3$ holds in A . So,

$$A \cong A/2A \times A/3A$$

by the Chinese Remainder Theorem. The factor $A/2A$ is a 3-ring with $2 = 0$, hence a 2-ring by Example 4.5 and therefore commutative. The factor $A/3A$ is a 3-ring with $3 = 0$, hence commutative by Theorem 5.5. So A is a product of two commutative rings, and we are done.

Just by unwinding what happens in the proofs, the equational proof looks as follows: It is sufficient to show $xy \equiv yx \pmod{2}$ and $xy \equiv yx \pmod{3}$, since then $xy \equiv yx \pmod{6}$ by the proof in Theorem 3.14 and hence $xy = yx$. In fact, if $xy - yx = 2u$ and $xy - yx = 3v$, then $xy - yx = 6(u - v) = 0$. We have

$$x + 1 = (x + 1)^3 = x^3 + 3x^2 + 3x + 1 = 3x^2 + 4x + 1 \equiv x^2 + 1 \pmod{2}$$

and hence $x^2 \equiv x \pmod{2}$ for all x . Then (we now repeat Example 2.4)

$$x + y \equiv (x + y)^2 = x^2 + xy + yx + y^2 \equiv x + xy - yx + y \pmod{2}$$

and hence $xy \equiv yx \pmod{2}$. Now for the modulo 3 part: We have

$$1 \cdot (1 - (x - 1)^2) + 2 \cdot (1 - (x - 2)^2) = -3x^2 + 10x - 6 \equiv x \pmod{3}.$$

So it suffices to show that $xy^2 \equiv y^2x \pmod{3}$ for all x, y . But y^2 is central by Lemma 2.8 and Lemma 2.9 (we don't need to repeat the proofs here).

Of course, there are simpler proofs of the commutativity of 3-rings (such as the one in Example 2.11), but the advantage is that our proof has been produced with a *general method*. It was not necessary to come up with clever ideas, and the same method works for larger numbers as well, as long as they are simple. Let us also mention that the reduction to p -rings will often be rather simple because of Remark 4.17.

Example 5.9. Let us show with our general method that every 5-ring is commutative. We already saw in Example 3.15 that it is enough to prove this modulo the primes 2, 3, 5. For the primes 2 and 3, Examples 4.6 and 4.7 reduce the problem to 2- and 3-rings, which we have covered before. For the prime 5, we can use Theorem 5.5. Explicitly, the equation

$$1 \cdot (1 - (x - 1)^4) + 2 \cdot (1 - (x - 2)^4) + 3 \cdot (1 - (x - 3)^4) + 4 \cdot (1 - (x - 4)^4) \equiv x \pmod{5}$$

shows that x is central modulo 5 because each y^4 is central by Lemmas 2.8 and 2.9.

Example 5.10. Let us show with our general method that every 21-ring is commutative. The primes p with $p - 1 \mid 21 - 1$ are $p = 2, 3, 5, 11$, we must have $2 \cdot 3 \cdot 5 \cdot 11 = 0$ in any 21-ring: This is because $2 \cdot 3 \cdot 5 \cdot 11 = 330 = \gcd(2^{21} - 2, 3^{21} - 3, 5^{21} - 5)$. Thus, it suffices to show commutativity modulo each of these primes. For $p \in \{2, 3, 11\}$ we find

$$\gcd(T^{21} - T, (T + 1)^{21} - (T + 1)) = T^p - T$$

in $\mathbb{F}_p[T]$, showing that every 21-ring with $p = 0$ is a p -ring and hence commutative by Theorem 5.5. In $\mathbb{F}_5[T]$ we find

$$\gcd((T + u)^{21} - (T + u) : u \in \mathbb{F}_4) = T^5 - T,$$

showing that every 21-ring with $5 = 0$ is a 5-ring and hence commutative by Theorem 5.5.

The reduction method can be applied to non-simple numbers as well. Let us look at some examples. More results of this type are summarized in Table 1.

Example 5.11. Let us show with our general method that every 7-ring is commutative, even though 7 is not simple. The primes p with $p - 1 \mid 7 - 1$ are $p = 2, 3, 7$, and since $2 \cdot 3 \cdot 7 = 42 = \gcd(2^7 - 2, 3^7 - 3)$, it suffices to prove commutativity modulo each of these primes. We already saw in Example 4.13 that a 7-ring with $2 = 0$ is a 4-ring, hence commutative by Example 2.12. In $\mathbb{F}_3[T]$ we find

$$\gcd(T^7 - T, (T + 1)^7 - (T + 1)) = T^3 - T,$$

showing that every 7-ring with $3 = 0$ is a 3-ring and hence commutative. Finally, every 7-ring with $7 = 0$ is commutative by Theorem 5.5.

Example 5.12. Let us show with our general method that every 2023-ring is commutative. The primes p with $p - 1 \mid 2023 - 1$ are $p = 2, 3, 7$, and since $2 \cdot 3 \cdot 7 = 42 = \gcd(2^{2023} - 2, 3^{2023} - 3)$, it suffices to prove commutativity modulo each of these primes. In $\mathbb{F}_2[T]$ we find

$$\gcd(T^{2023} - T, (T + 1)^{2023} - (T + 1), (T^2 + T + 1)^{2023} - (T^2 + T + 1)) = T^4 - T,$$

so that every 2023-ring with $2 = 0$ is a 4-ring and hence commutative. In $\mathbb{F}_3[T]$ we find

$$\gcd(T^{2023} - T, (T + 1)^{2023} - (T + 1), (T + 2)^{2023} - (T + 2)) = T^3 - T,$$